

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Electrical Engineering

2. Subject Name: Digital Electronic Circuits Laboratory, L-T-P: 0-0-3, Credits: 2

3. Pre-requisites: No

4. Syllabus and reference books:

Syllabus:

Experiment 1: Familiarization with 7-segment (common-anode) light-emitting diode (LED) display and IC 7447 (BCD-to-7-segment decoder/driver)

Experiment 2: Familiarization with 7493 (mod-16 counter) and 7490 (mod-10 counter)

Experiment 3: Familiarization with IC 74181 (4-bit ALU) chip

Experiment 4: To verify the truth table of the 4-bit adder IC 7483, and using quad 2-to-1 Multiplexer IC 74157, display each of the 4-bit inputs and the output, one at a time by a 7-segment LED display

Experiment 5: Use IC 74173 (quad DFF) and IC 74374 (octal DFF) to build a 1-digit calculator

Experiment 6: Implementation of a (4×4) -bit multiplier using registers and a down-counter. Use principles of repeated addition

Experiment 7: BCD addition (add two BCD numbers to produce a valid BCD result)

Experiment 8: Repetitive display of a student's roll number on 7-segment display

Experiment 9: Simulate a serial communication link. Assume RS-232 data format consisting of a low start bit, 4 data bits, a parity bit and 2 high stop bits.

Experiment 10: Mini project on electronic system design

Reference Books:

1) "Digital Design", M. Morris Mano and Michael D. Ciletti

2) "Digital Integrated Circuits", Jan M. Rabaey, Anantha Chanrakasan and Borivoje Nikolic

3) "Foundation of Analog and Digital Electronic Circuits", Anant Agarwal and Jeffrey Lang

4) "Digital Fundamentals", Thomas L. Floyd

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Familiarization with 7-segment (common-anode) light-emitting diode (LED) display and IC 7447 (BCD-to-7-segment decoder/driver)	3 hours
2.	Familiarization with 7493 (mod-16 counter) and 7490 (mod-10 counter)	3 hours
3.	Familiarization with IC 74181 (4-bit ALU) chip	3 hours
4.	To verify the truth table of the 4-bit adder IC 7483, and using quad 2-to-1 Multiplexer IC 74157, display each of the 4-bit inputs and the output, one at a time by a 7-segment LED display	3 hours
5.	Use IC 74173 (quad DFF) and IC 74374 (octal DFF) to build a 1-digit calculator	3 hours
6.	Implementation of a (4×4) -bit multiplier using registers and a down-counter. Use principles of repeated addition	3 hours
7.	BCD addition (add two BCD numbers to produce a valid BCD result)	3 hours
8.	Repetitive display of a student's roll number on 7-segment display	3 hours
9.	Simulate a serial communication link. Assume RS-232 data format consisting of a low start bit, 4 data bits, a parity bit and 2 high stop bits.	3 hours
10.	Mini project on electronic system design	9 hours
Total number of hours		36 hours